MLP Project Milestone 1: Exploratory Data Analysis

To understand about all the milestones in detail, Please *click here*

Roll number	

Submission guidelines

1. **Make your own copy with File > Make a copy, and write only in your copy.**
2. **Once you have finished writing your report, save or export the .docx document as a PDF.**
3. **Rename the PDF file using the following exact convention: [roll-number]_26t3_M[Milestone-Number].pdf**
4. **Example: If your roll number is 21f1001234 and you are submitting the PDF for Milestone 1, your file must be named exactly: 21f1001234_26t3_M1.pdf**

Instructions

1. Write each answer in the "Your answer" box under the question, and paste the Python code (wherever required) you ran into the "Your code" box as text, not as a screenshot.
2. Support every claim with evidence from your own analysis of the competition data: a number, a small table or a plot (Insert > Image). Summarise long outputs instead of pasting them in full.
3. The points under each question describe what an answer should cover. If a point does not apply to your dataset (for example, there are no date columns), you can skip it.
4. Write in your own words. Clear reasoning backed by evidence matters more than length of your answer.
5. There is no single correct answer, these questions are subjective and there can be multiple valid explanations of questions. Use your interpretation and understanding to answer the questions.
6. You need not include all the columns(features) while answering questions, you can select a few meaningful columns and do analysis on them. Try to cover at least 50% of the columns(features) for the questions in this milestone.

7. You MAY use a different kaggle notebook instead of the one shared with the host to do the task given in milestones.
8. You do not need to analyse all the columns;however, make sure to cover a good number of columns to support your analysis and decisions. Complete EDA need not be covered during this milestone, you can do more on EDA during the next milestones if required.
9. The answer length can vary depending on questions. Keep your answer concise and give a summary of the analysis and cover details using visualizations/bullet points wherever required. The maximum length of answer can be considered around 100 to 400 words.

Q1. Problem statement

What problem does this project ask you to solve, and what type of machine learning problem is it? Explain your understanding in your own words.

One suggested approach would be as follows:

- The problem in your own words: what is being predicted, and what a single prediction is about (for example one customer, one transaction or one product).

Your answer

Write your answer here.

Q2. Dataset structure, schema and data types

Describe the given dataset. Describe the files, what a single row represents, and the role and data type of columns.

One suggested approach would be as follows:

- The files provided and what each is for; the shape (rows × columns) of the training and test data; and any column present in one file but not the other, with the reason.
- The data types of columns(features).

Your answer

Write your answer here.

Your code

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# Paste the code you ran for this question
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Q3. Target variable and evaluation metric

What does the target look like, and what does the evaluation metric used in competition capture?

One suggested approach would be as follows:

- The distribution of the target in the training data.
- Understanding of evaluation metric used in competition.

Your answer

Write your answer here.

Your code

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# Paste the code you ran for this question
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Q4. Missing values

How much data is missing, where, and why? Decide whether the missingness is random or structured, whether it says anything about the target, and how you will handle it.

One suggested approach would be as follows:

- Analysis of missing data.
- Process of identifying missing values, types of missing values.
- Any pattern captured in missing values.

Your answer

Write your answer here.

Your code

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# Paste the code you ran for this question
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Q5. Data integrity: duplicates, invalid and inconsistent values

Which records or values are duplicated, impossible, inconsistent? For each problem, decide whether it is an error or a genuine but unusual case, and say what you will do about it.

One suggested approach would be as follows:

- Analysis of duplicate records.
- Process/approach of finding/deciding importance of features.
- Less important columns as per your understanding.
- For each issue: whether it is an error or a genuine value, and your action with the reason.

Your answer

Write your answer here.

Your code

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# Paste the code you ran for this question
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Q6. Numerical features: distributions, outliers and scale

What do the distributions of the numerical features look like? Describe their shapes, scales and outliers, and explain which transformations, if any, they need.

One suggested approach would be as follows:

- Summary statistics of some important columns as per your understanding/interpretation.
- Distribution of values of the columns.
- Findings from various methods used for analysing columns as per your decision/choice.

Your answer

Write your answer here.

Your code

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# Paste the code you ran for this question
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Q7. Categorical and other non-numeric features

What do the categorical and other non-numeric columns contain and how should each be represented for a model of your choice?

One suggested approach would be as follows:

- Types of values and their frequency distributions.
- Handling of various types of categorical columns(features).
- Other special columns, if any: the appropriate analysis for these columns.

Your answer

Write your answer here.

Your code

```
# Paste the code you ran for this question
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Q8. Relationships between features and the target

Which features carry the most information about the target, and what form does each important relationship take?

One suggested approach would be as follows:

- Your expectations before looking: from the problem description and common sense, which features should matter and in which direction?
- A ranked list of roughly the ten most informative features under a measure you state.
- Statistics or plots to find patterns/relationships.

Your answer

Write your answer here.

Your code

```
# Paste the code you ran for this question
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Q9. Relationships among features: correlation, redundancy and interactions

How are the features related to one another? Identify groups of redundant or strongly related features. Describe how the relationship between features impact the choice of models?

One suggested approach would be as follows:

- Analysis to extract patterns/relation among columns(features).
- Associations between categorical and numeric features, if any.

Your answer

Write your answer here.

Your code

```
# Paste the code you ran for this question
```

Q10. Key insights and preprocessing plan

Tell your top 3 findings. Which insights matter most, and what exactly will you do in the next milestones because of them?

One suggested approach would be as follows:

- A preprocessing plan for each group of features.
- At least three feature-engineering ideas that come from your EDA.
- The model families you expect to do well or poorly given the characteristics of this data, and the simple baseline model you will start from.
- The main risks if they are present and how you will monitor or resolve them.

Your answer

Write your answer here.

Your code

```
# Paste any code you ran for this question
```